

- Q.21 Describe the importance of cylinder cushioning in linear actuators. (CO4)
- Q.22 How does a meter-in circuit function in controlling the speed of a hydraulic cylinder? (CO5)

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x8=16)
- Q.23 Explain the working and applications of hydraulic rotary actuators. (CO4)
- Q.24 Design a fluid power circuit for directional control of a double-acting hydraulic cylinder using lever operated valves. (CO5)
- Q.25 Discuss the types of air compressors used in pneumatic power supplies. Explain any one with neat sketch. (CO2)

No. of Printed Pages : 4
Roll No.

222844/212844

4th Sem. / Automation and Robotics

Subject : Fluid Power Systems

Time : 3 Hrs.

M.M. : 60

SECTION-A

- Note:** Multiple choice questions. All questions are compulsory (6x1=6)
- Q.1 Which among the following are not the main selection criterion for selection of hydraulic pumps?
- a) Discharge b) Pressure (CO2)
 - c) Speed d) Weight
- Q.2 Pascal's law is applied in which of the following?
- a) Electrical Circuits (CO1)
 - b) Fluid Power Systems
 - c) Thermal Systems
 - d) None of the above
- Q.3 A single-acting pneumatic cylinder operates using a:
- a) 4/2 Directional Control Valve (CO5)
 - b) 3/2 Directional Control Valve
 - c) 5/3 Directional Control Valve
 - d) Pressure Relief Valve

- Q.4 What is the main function of an accumulator in a hydraulic system? (CO2)
- Store electrical energy
 - Control direction of flow
 - Store hydraulic fluid under pressure
 - Lubricate the pump
- Q.5 A poppet valve is classified as which type of valve?
- Rotary Valve
 - Spool Valve
 - Proportional Valve
 - Seat Valve
- Q.6 Which type of motion is transmitted by hydraulic actuators? (CO4)
- Linear motion
 - Rotary motion
 - Both a and b
 - None of the above

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 The unit of pressure is _____. (CO1)
- Q.8 Hydraulic system use _____ as a working fluid. (compressible/Incompressible fluid) (CO2)
- Q.9 A quick-exhaust valve is used to _____. (CO3)
- Q.10 Linear actuators are typically used for _____ movement. (CO4)

- Q.11 A meter-out circuit is used to control the _____ of the actuator. (Inflow/Outflow) (CO5)
- Q.12 A shuttle valve is used in fluid power systems for _____ purposes. (CO3)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Explain the main characteristics of industrial prime movers. (CO1)
- Q.14 Describe the components of a hydraulic power supply. (CO2)
- Q.15 Differentiate between single-acting and double-acting cylinders. (CO4)
- Q.16 How does a 4/2 directional control valve work in a hydraulic system? Explain with neat sketch. (CO3)
- Q.17 What is the principle of Pascal's Law and how is it applied in fluid power systems? (CO1)
- Q.18 Discuss the construction of a spool valve with neat sketch. (CO3)
- Q.19 Explain the sequencing operation of two pneumatic cylinders using pressure-based sequencing valves. (CO5)
- Q.20 Classify different types of hydraulic pumps? (CO2)